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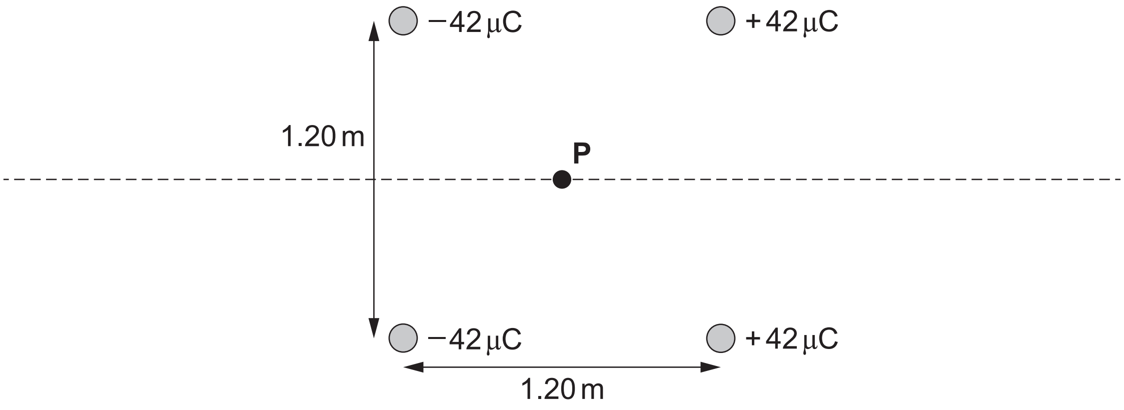
4. Four point charges are placed at the corners of a square of length 1.20 m as shown. Point **P** is located at the centre of the square.



(a) (i) Define electric field strength. [1]

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(ii) Show that the magnitude of the electric field strength at **P** due to any one of the four charges is approximately 520 kN C^{-1}. [3]

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(iii) Hence, calculate the resultant electric field strength at **P** and give its direction. [3]

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(b) A negative point charge is held at **P** and is free to move along the dotted line shown in the diagram with no resistive forces.

(i) Define electric potential. [1]

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(ii) Explain briefly why the potential energy of the negative charge is zero at **P**. [2]

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(iii) Aled states "The negative charge initially accelerates to the right, then slows down but will just about reach infinity if no resistive forces act." Discuss to what extent Aled's points are correct. [3]

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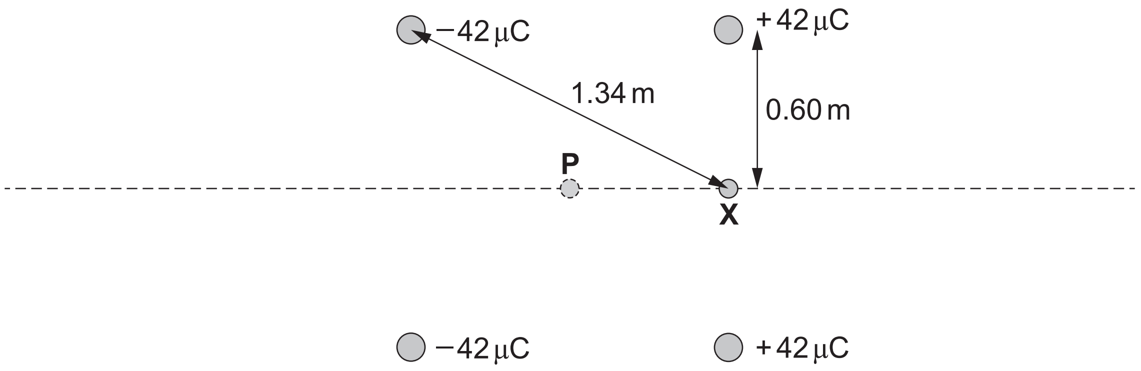
(iv) A charge of −38 nC and mass of 7.7×10^{-4} kg is released from rest from point **P**. Calculate the speed of the −38 nC charge when it reaches point **X** shown below. [5]



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